



Artificial intelligence adoption and corporate financial risk

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ABSTRACT

Artificial intelligence (AI) is a critical driver of competitive advantage. Using data from China's A-share listed companies (2010–2022), this study finds that AI adoption significantly reduces corporate financial risk through structural, resource-based, and technological channels—especially in nonhigh-tech firms and industries with lower uncertainty.

1. Introduction

Rapid digital advancement and complexity present firms with both challenges and opportunities. Hence, companies must innovate to improve quality, reduce costs, increase efficiency, and strengthen user engagement (Igna and Venturini, 2023; Babina et al., 2024; Zhang et al., 2025). In this context, artificial intelligence (AI) is a transformative force in building value propositions and unlocking opportunities (Mustak et al., 2021; Czarnitzki et al., 2023; Zhao et al., 2025). Corporate AI adoption rose from 20 % in 2017 to 50 % in 2022, with most firms allocating >5 % of their budgets to AI initiatives. Moreover, about 23 % applied at least 5 % of their 2022 earnings before interest and taxes AI.

The literature documents various business impacts of AI adoption. It lowers production costs, boosts efficiency, and enables cost-leading firms to earn excess profits (Bosse et al., 2023; Wu et al., 2025). AI automates traditional production inputs, reduces labor needs and defect rates, and improves product quality. Integrated with technologies such as the Internet of Things and cloud computing, AI supports intelligent manufacturing (Li et al., 2023), shortens development cycles, and raises innovation efficiency (Bahoo et al., 2023). However, AI's potential for mitigating corporate risk remains underexplored.

Traditional risk models, dependent on historical data and human judgment, often fail to respond to dynamic markets and are prone to error (Kemp, 2024). AI enables real-time monitoring of financial health, better risk forecasting, and smarter decision-making, all of which reduce financial risk. Tools like robotic process automation and intelligent data analysis further limit operational mistakes and improve efficiency, bolstering resilience. Assessing AI's influence on corporate financial risk is therefore critical. This study analyzes data from listed firms between 2010 and 2022 to evaluate AI's impact on financial risk amid a complex economy. Findings show AI enhances corporate risk resistance through structural, resource, and technological channels, especially in nonhigh-tech sectors and industries with lower uncertainty.

Although prior research has focused on AI's role in intelligent manufacturing and human resource (HR), few have explored its function in risk resilience. We investigate AI's importance in this area, demonstrating its benefits under specific conditions, and present empirical evidence on how it reinforces financial risk management. Second, we examine these effects in complex economic conditions, providing a theoretical framework explaining how emerging technologies transform risk management.

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Table 1
Variable definitions.

Variable	Definition	Measure
Corporate financial risk	<i>Zscore</i>	Z-score model.
Application of artificial intelligence	<i>AI</i>	Permeability of industrial robots
Firm size	<i>Size</i>	Ln (total assets)
Firm age	<i>FirmAge</i>	Ln(#Years of incorporation)
Leverage ratio	<i>Lev</i>	Total debt/Total assets
Return on assets	<i>ROA</i>	Net profit/Average net assets
Cash flows	<i>Cashflow</i>	Cash flow/Total assets
Board size	<i>Board</i>	Ln(#Directors at the end of the year)
Proportion of independent directors	<i>Indep</i>	#Independent directors/#Board members
Largest shareholder's shareholding ratio	<i>Top1</i>	Largest shareholder's shareholding/Total share capital
Combining the positions of chairman and chief executive	<i>Dual</i>	Binary = 1 If the chairman and chief executive is the same person; zero otherwise

2. Research hypothesis

AI supports innovation, cost-cutting, and improved efficiency by optimizing HR management, increasing skilled labor, and improving structural stability (von Krogh et al., 2023; Benedetti et al., 2025). These functions help reduce risks linked to staff turnover and talent shortages. AI-powered recruitment and training tools are used to attract and retain top talent while easing financing constraints through precise assessments and credit scoring.

By applying AI to big data analysis, firms can better demonstrate repayment ability and growth prospects to financial institutions (Goto, 2023; Jia, 2025), boosting credibility and attracting capital. Intelligent financial tools enhance cash flow forecasting and budgeting, increasing financial resilience. AI also automates workflows and enables smarter decisions, improving both productivity (Nucci et al., 2023; Lai et al., 2024) and profitability. These systems speed up operations, lower errors, and raise efficiency (Damioli et al., 2023). Deep learning and data mining reveal trends in market demand and supply chains, supporting proactive strategies that cut resource waste. This improved productivity sharpens competitiveness and expands profit margins, key to managing economic fluctuations and uncertainty. Based on this analysis, this paper proposes Hypothesis 1:

H1. The application of AI technology reduces corporate financial risks.

3. Research design

3.1. Sample selection

This study analyzed Chinese A-share listed firms from 2010 to 2022, using data from the CSMAR Database. After excluding ST firms and those with major data gaps, the final unbalanced panel included 3530 firms and 27,690 observations. To reduce outlier influence, all continuous variables were winsorized at the 1 % level on both tails of the distribution.

3.2. Variable definition

Corporate financial risk was measured using Altman's (1968) Z-score model, where a higher value indicates lower financial risk. Thus, firms with higher Z-scores face lower financial risk (*Zscore*).

AI adoption (*AI*) was assessed using industrial robot penetration, following Du et al. (2024). The industry-level robot penetration index is calculated as the ratio of robot stock to total employees in 2010. The firm-level penetration is then derived by multiplying the 2011 share of production department employees by the industry median for that year and the industry-level index, yielding each firm's AI usage level.

Control variables, drawn from Zhao et al. (2024), included firm size, age, debt-to-asset ratio, return on assets, cash flow, board size, share of independent directors, ownership share of the largest shareholder, and chief executive–chairman duality. Table 1 provides detailed definitions of these variables.

3.3. Model construction

To assess the effect of AI on financial risk, we use the following model:

$$Zscore_{it} = \alpha_0 + \alpha_1 AI_{it} + \alpha_2 Controls_{it} + \gamma_i + \varphi_t + \varepsilon_{it} \quad (1)$$

where *Zscore* is the dependent variable measuring financial risk for firm *i* in year *t*; *AI* is the main independent variable capturing AI adoption; *Controls* is the vector of control variables.

Table 2
Descriptive statistics.

Variables	N	Mean	SD	Min	Max
<i>Zscore</i>	27,690	4.987	5.366	0.297	36.00
<i>AI</i>	27,690	6.809	4.045	0.138	15.12
<i>Size</i>	27,690	22.26	1.275	20.13	26.35
<i>FirmAge</i>	27,690	2.897	0.333	1.792	3.526
<i>Lev</i>	27,690	0.409	0.194	0.055	0.843
<i>ROA</i>	27,690	0.045	0.059	−0.181	0.222
<i>Cashflow</i>	27,690	0.050	0.065	−0.134	0.237
<i>Board</i>	27,690	2.123	0.198	1.609	2.708
<i>Indep</i>	27,690	0.376	0.053	0.333	0.571
<i>Top1</i>	27,690	0.338	0.147	0.084	0.741
<i>Dual</i>	27,690	0.293	0.455	0.000	1.000

Table 3
Baseline results.

Variable	(1) Zscore	(2) Zscore
<i>AI</i>	0.014*** (2.67)	0.012*** (2.67)
<i>Size</i>		−0.673*** (−6.67)
<i>FirmAge</i>		0.567 (0.92)
<i>Lev</i>		−12.055*** (−28.52)
<i>ROA</i>		10.397*** (16.15)
<i>Cashflow</i>		2.073*** (5.31)
<i>Board</i>		−0.206 (−0.80)
<i>Indep</i>		0.103 (0.13)
<i>Top1</i>		−2.064*** (−4.17)
<i>Dual</i>		−0.239*** (−2.67)
Constant	6.542*** (48.02)	24.698*** (9.05)
Observations	27,690	27,690
R ²	0.093	0.272
Firm	Yes	Yes
Year	Yes	Yes

Note: *t*-statistics are in parentheses. * $p < 0.1$, ** $p < 0.05$.
*** $p < 0.01$.

4. Empirical results

4.1. Descriptive statistics

Table 2 presents descriptive statistics for the study's key variables. From 2010 to 2022, the average firm financial risk is 4.987, with a maximum of 36, indicating broad variability. The mean AI adoption score is 6.809, with a minimum of 0.138, reflecting significant differences in AI usage across listed firms. Other variables are consistent with prior research.

4.2. Baseline results

Table 3 displays baseline regression outcomes on AI's impact on financial risk. Column (1) shows the AI coefficient is significantly positive at the 1 % level, even without control variables. In Column (2), after adding controls, the AI coefficient remains significantly positive. A 1 % increase in AI usage corresponds to a 0.012 % drop in financial risk, confirming that AI adoption significantly mitigates financial risk and supports H1.

Table 4
Mechanism test results.

Variable	(1) Ratio	(2) SA	(3) TFP
<i>AI</i>	0.028** (2.22)	−0.000*** (−2.63)	0.001** (2.34)
Constant	20.067** (2.48)	−3.627*** (−31.30)	−2.680*** (−7.31)
Observations	27,690	27,690	27,690
R ²	0.074	0.762	0.472
Controls	Yes	Yes	Yes
Firm	Yes	Yes	Yes
Year	Yes	Yes	Yes

Note: *t*-statistics are in parentheses. * $p < 0.1$,.

** $p < 0.05$,.

*** $p < 0.01$.

Table 5
Main variable replacement results.

Variable	(1) Zscore	(2) Oscore
<i>AI</i>	0.015*** (4.19)	
<i>AI</i>		−0.005* (−1.92)
Constant	23.398*** (8.72)	−4.234*** (−4.28)
Observations	27,690	25,650
R ²	0.274	0.488
Controls	Yes	Yes
Firm	Yes	Yes
Year	Yes	Yes

Note: *t*-statistics are in parentheses.

* $p < 0.1$, ** $p < 0.05$,.

*** $p < 0.01$.

Table 6
Other robustness test results.

Variable	(1) Zscore	(2) Zscore	(3) PSM 1:1	(4) PSM radius
<i>AI</i>	0.011*** (2.61)		0.011** (2.34)	0.012*** (2.66)
<i>L.AI</i>		0.008* (1.68)		
Constant	23.277*** (7.89)	23.281*** (7.92)	25.087*** (9.13)	24.795*** (9.10)
Observations	27,690	23,758	26,433	27,674
R ²	0.337	0.278	0.272	0.272
Controls	Yes	Yes	Yes	Yes
Firm	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
Year*Ind	Yes	No	No	No

Note: *t*-statistics are in parentheses.

* $p < 0.1$.

** $p < 0.05$.

*** $p < 0.01$.

4.3. Mechanism tests

The theoretical analysis suggests AI adoption may lower financial risk through structural, resource-based, and technological channels. To verify this, we followed [Bernardi and Arnold \(1997\)](#), [Ren et al. \(2025\)](#), and [Battisti et al. \(2020\)](#), using proxies for each mechanism: the proportion of high-end talent (structural), financing constraints (size–age index; resource-based), and total factor productivity (TFP), using the Levinsohn–Petrin model.

Table 7
Instrumental variable test results.

Variables	(1) AI	(2) Zscore
AI-City	0.044*** (4.61)	
AI		1.9875*** (4.15)
Constant	4.821*** (7.78)	27.4544*** (10.22)
Observations	27,690	27,690
Controls	Yes	Yes
Firm	Yes	Yes
Year	Yes	Yes
Weak IV test (F value)	68.47	

Note: t-statistics are in parentheses. * $p < 0.1$, ** $p < 0.05$.

*** $p < 0.01$.

Table 4 supports these mechanisms. Column (1) shows AI significantly boosts recruitment of high-quality talent (structural effect). Column (2) reveals that AI helps firms overcome capital market financing barriers (resource effect). Column (3) confirms that AI adoption enhances TFP (technological effect). Together, these results indicate that AI reduces financial risk via these three pathways.

4.4. Robustness test

4.4.1. Replacement of main variables

Following Zhong et al. (2024), we used text analysis to quantify AI-related keywords in listed firms' annual reports. The values are adjusted by adding one and applying a natural logarithm, creating an alternative AI adoption metric (*AI1*). In Table 5, Column (1), the coefficient for *AI1* is 0.015 and significant at the 1 % level, affirming the robustness of the findings.

To strengthen the analysis, financial risk was remeasured using the Oscore metric, where higher scores indicate greater risk. Column (2) of Table 5 shows that results remain significant, supporting the conclusion that AI adoption reduces financial risk across different risk measures.

4.4.2. Controlling for more fixed effects

Although the regressions control for year and firm fixed effects, they may overlook industry-specific trends. To address this, year-industry interaction fixed effects were added. Table 6, Column (1), confirms that the results remain robust after accounting for industry fluctuations.

4.4.3. Considering lag effects

To mitigate potential endogeneity, the independent variable is lagged by one period. Results in Table 6, Column (2), confirm the model's stability and reinforce that AI adoption significantly reduces corporate financial risk.

4.4.4. Propensity score matching (PSM)

To address sample selection bias, this study used PSM for robustness checks. Firms are grouped by average AI adoption: those above the mean form the treatment group; the rest form the control group. Both 1:1 nearest-neighbor and radius matching are applied, using control variables as covariates.

Regression results in Table 6, Columns (3) and (4), confirm consistency and robustness, supporting the finding that AI adoption significantly lowers corporate financial risk.

4.4.5. Instrumental variable

As an instrumental variable for AI development, the total number of new AI firms established in a city is used. This count is positively correlated with firm-level AI adoption but unlikely to directly influence individual firms' financial risk, meeting the requirements of a valid instrument.

Table 7 shows a significant relationship between the instrument and firm-level AI use, with a coefficient of 0.044 (significant at the 5 % level). Column (2) indicates that, after addressing endogeneity, the AI coefficient on financial risk is 1.9875, significant at the 1 % level. These results further validate the robustness of the study's conclusions.

4.5. Heterogeneity effects

Nonhigh-tech firms typically rely on traditional management systems and simpler infrastructure, making them less dependent on advanced algorithms (Gharbi et al., 2014). Adopting AI (e.g., robotic process automation or predictive analytics) can yield substantial cost savings and efficiency gains, reducing operational and liquidity risks. As a result, these firms experience stronger marginal benefits from AI.

Table 8
Heterogeneity effects.

Variables	(1) High_tech	(2) Non-High_tech	(3) High_uncertainty	(4) Low_uncertainty
AI	0.010 (1.50)	0.015*** (2.60)	0.007 (1.05)	0.014** (2.57)
Constant	37.123*** (9.03)	11.958*** (3.57)	26.687*** (7.72)	28.209*** (6.34)
Observations	15,464	12,226	13,929	13,761
R ²	0.304	0.236	0.267	0.303
Controls	Yes	Yes	Yes	Yes
Firm	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes

Note: t-statistics are in parentheses. * $p < 0.1$.

** $p < 0.05$.

*** $p < 0.01$.

In low-uncertainty industries, firms operate in stable conditions with less market volatility and competitive pressure (Wang et al., 2023; Wei et al., 2024). Such environments allow more focus on internal process optimization without frequent strategic shifts. AI strengthens planning, early risk detection, and preventive action. Thus, in these settings, AI's risk-mitigation impact is amplified, maximizing its value in enhancing decision-making and operational efficiency.

This study examined AI's heterogeneous effects on corporate financial risk by dividing the sample into high-tech (*High_tech*) and nonhigh-tech (*Non-High_tech*) groups, based on the Guidelines for Industry Classification of Listed Companies (2012 Revision). Table 8, Columns (1) and (2), shows that AI more effectively reduces financial risk in nonhigh-tech firms. Industry uncertainty is measured by the median unadjusted environmental uncertainty of firms within the same industry and year. Based on the mean, industries are categorized into high (*High_uncertainty*) and low (*Low_uncertainty*) uncertainty groups. Columns (3) and (4) of Table 8 indicate that AI's risk-reducing effect is more pronounced in low-uncertainty industries.

5. Conclusion

Amid rapid advances in information technology, AI has reshaped corporate operations across sectors. Using data from China's A-share listed firms from 2010 to 2022, this study investigated how AI adoption influences corporate financial risk. Results show that AI enhances talent acquisition, reduces financing costs, and boosts productivity, substantially lowering financial risk. This effect is particularly strong in nonhigh-tech firms and industries with low environmental uncertainty. The findings expand the literature on AI's economic impact and deepen understanding of corporate financial risk in an era of digital transformation.

Author statement

This research investigates the impact of artificial intelligence (AI) adoption on corporate financial risk, utilizing data from China's A-share listed companies between 2010 and 2022. The study highlights how AI reduces financial risks through structural, resource, and technological mechanisms, with a particular focus on its differential effects in non-high-tech firms and industries with lower environmental uncertainty.

CRedit authorship contribution statement

Shuai Liu: Writing – original draft, Supervision, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Lihui Gao:** Supervision, Resources, Methodology, Investigation, Data curation. **Mengzhu Chen:** Writing – review & editing, Writing – original draft, Software, Resources, Methodology, Investigation.

Declaration of interest

None.

Data availability

Data will be made available on request.

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